Current Affairs 12-11-2026

Credit - Drishti Ias

DPIIT’s Push to Indigenise 100 Critical Items

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For Prelims: Department for Promotion of Industry and Internal Trade , Critical minerals , Minimum Import Price , India Semiconductor Mission , KABIL , Minerals Security Partnership , Indo-Pacific Economic Framework

For Mains: India's bilateral relations with China, Balancing strategic alignment with Economic nationalism

Source: IE

Why in News?

The **Department for Promotion of Industry and Internal Trade (DPIIT)** has spearheaded a strategic drive to **indigenise the domestic production** of **100 critical intermediates and capital goods** .

- This intervention comes in the immediate backdrop of **China’s Ministry of Commerce** announcing an aggressive export licensing regime, effective 18 th November 2026, targeting specialized capital goods such as tunnel boring machines and upstream solar manufacturing technologies.

Summary

- DPIIT's 100-item indigenisation initiative aims to reduce India's dependence on Chinese imports, strengthen domestic capital goods manufacturing and enhance supply-chain resilience.
- Addressing technological gaps, high production costs and critical mineral dependence through R&D, overseas acquisitions and policy reforms is essential to achieve industrial self-reliance and strategic autonomy.

What is Driving DPIIT's Indigenisation Push?

- **Supply Chain Weaponisation:** China is actively weaponising its manufacturing monopoly, evidenced by its upcoming November 2026 export licensing curbs on specialized capital machinery.
  - Just as China's 2023 export curbs on **critical minerals (Gallium, Germanium, and Graphite)**

exposed global tech vulnerabilities, indigenising heavy machinery ensures **India's infrastructure and renewable transitions remain completely insulated from similar geopolitical coercion.**

- **Unsustainable Trade Asymmetry:** Despite India imposing reactive anti-dumping duties, the **bilateral trade deficit with China crossed a staggering USD 100 billion in FY 2025-26** .
  - This bilateral imbalance reflects a much larger global trend. Driven by a massive surge in high-tech exports, China logged a USD 1.04 trillion global trade surplus between January - October 2026. This puts Beijing on track to surpass its record USD 1.2 trillion surplus in 2025.
  - This targeted indigenisation push acts as a structural corrective mechanism, **driving aggressive import substitution to arrest this massive drain of foreign exchange reserves.**
- **Countering Global Export Monopolies** : A Rhodium Group report projects that China's global trade surplus will reach **USD 1.2 trillion in 2025, with an additional 8% share of global trade by 2030.**
  - Drawing lessons from **India's smartphone manufacturing success** , which reduced import dependence from 78% in 2014 to becoming the **world's second-largest mobile phone manufacturer,** indigenising critical industrial intermediates can build domestic economies of scale, reduce import dependence and **position India as a competitive net exporter under the China Plus One strategy, particularly in heavy industries.**
- **Escaping the 'Assembly Trap':** India's previous manufacturing successes have largely been restricted to the downstream assembly of imported components **(Semi-Knocked Down kits).**
  - This drive pivots toward upstream resilience, fostering deep-tech capabilities and a self-sustaining industrial ecosystem.

Which are the Key Sectors Targeted in the DPIIT 100-Item List?

- **Six Sector-Specific Working Groups:** The government constituted six working groups covering:
  - Pharmaceuticals, biotechnology and medical devices.
  - Chemicals, petrochemicals, textiles and footwear.
  - Capital goods, automobiles and electric vehicles.
  - Energy.
  - Construction equipment and infrastructure.
  - Defence and aerospace (civilian-use items) and electronics.
- **Solar Manufacturing (Bridging the Upstream Void):** India boasts a massive downstream **solar module assembly capacity of 233 GW** annually.
  - However, upstream capabilities are critically deficient - **solar cell capacity is only ~30 GW** , and the foundational **ingot and wafer manufacturing capacity stands at a mere 2 GW** .
  - The indigenisation push aims to shield new cell and ingot-wafer plants from Chinese capital goods restrictions. Simultaneously, the **Ministry of New and Renewable Energy (MNRE)** is formulating strict guidelines for **recycling solar module waste** to secure secondary raw materials.
- **Heavy Machinery and Capital Goods:** A primary target is the domestic manufacturing of **Tunnel Boring Machines (TBMs)** , which are indispensable for defense infrastructure, hydropower projects, and urban metro networks.
  - Anticipating Chinese curbs, a new TBM manufacturing plant has already been commissioned in Tamil Nadu.
- **Core Petrochemicals and Basic Chemicals:** Targeting the backbone of the pharmaceutical, textile,

packaging, and agricultural (urea) industries, the list includes **Polyvinyl Chloride (PVC), polyethylene, polypropylene, polystyrene, and ABS** .

- It also targets high-value chemical imports like **phosphoric acid, ammonia, acetic acid, and toluene** . Corporate giants are responding.
- **Reliance Industries (which holds 51% of domestic PVC capacity)** is scaling up to 1.5 MMT by 2026-27, while the Adani Group is establishing a new PVC facility in Mundra.

What are the Challenges Associated with Indigenisation Drive?

- **The "China Premium" and CAPEX Inflation:** Sourcing alternative capital goods and technology from the **EU, South Korea, or Japan is highly expensive** .
  - Industry estimates indicate this can **push up capital costs by 20% to 40%** , inflating the cost of domestic infrastructure and rendering Indian exports uncompetitive.
- **Technological and Process Know-How Deficit:** Establishing physical plants is only half the battle.
  - Acquiring proprietary engineering blueprints and achieving the high-efficiency process yields that Chinese manufacturers have perfected over decades remains a formidable barrier.
- **Risk of Monopolies and MSME Squeeze:** To shield domestic manufacturers, the government deployed protectionist tools like the **Minimum Import Price (MIP) of USD 0.766/kg on PVC Suspension Resin** (August 2026).
  - While this protects large conglomerates, it artificially inflates input costs for downstream MSMEs, shrinking their profit margins.
- **Gestation Lags:** Deep-tech capital goods manufacturing requires a gestation period of 3 to 5 years, leaving the Indian economy vulnerable to supply shocks in the interim.

What Measures Can India Adopt to Promote Indigenisation of Capital Goods?

- **Establish a National Capital Goods Mission (NCGM):** Modelled on the highly successful **India Semiconductor Mission (ISM)** , the government should launch a dedicated, empowered mission under the **Ministry of Heavy Industries** to oversee the 100-item list.
- **'Minelateralism' and Overseas Acquisitions:** To secure the foundational raw materials for capital goods, the government must aggressively expand the mandate and funding of **KABIL (Khanij Bidesh India Ltd.)** .
  - India must leverage diplomatic blocs like the **Minerals Security Partnership (MSP)** and the **Quad** to acquire equity in overseas critical mineral mines (e.g., in Africa, Australia, and Latin America).
  - India must actively utilize **Pillar II (Supply Chain Resilience)** of the **Indo-Pacific Economic Framework (IPEF)** and reinvigorate the **Supply Chain Resilience Initiative (SCRI)** with Japan and Australia.
- **Operationalizing the DESH Bill (2022):** Implementing the foundational recommendations of the **Baba Kalyani Committee (2018)** to overhaul the legacy **SEZ Act (2005)** via the proposed **Development of Enterprise and Service Hubs (DESH) Bill (2022)** .
  - This will create **WTO -compliant, mega-manufacturing enclaves** equipped with plug-and-play infrastructure, drastically slashing the high operational expenses (OPEX) for energy-intensive capital goods industries.

- **Fixing the Inverted Duty Structure:** Acting on **NITI Aayog** recommendations, the Finance Ministry must urgently rationalize customs frameworks. Raw materials and sub-components must attract the lowest tariffs, while a graded tariff structure should apply to finished capital goods to naturally incentivize domestic assembly.
- **Integrating the Circular Economy:** As India ramps up petrochemical and solar manufacturing, enforcing **Extended Producer Responsibility (EPR)** and urban mining will be crucial to mitigate environmental degradation and recover critical minerals.
- **Securing 'Patient Capital' and Boosting R&D:** **NaBFID** must deploy high-risk 'Patient Capital' and **Sovereign Green Bonds** to fund the long-gestation (3-5 years) capital goods projects that traditional commercial banks avoid.
    - Leverage the **Anusandhan National Research Foundation (ANRF) and the National Deep Tech Startup Policy (NDTSP)** to drive industry-academia R&D, securing indigenous patents in precision engineering.

Conclusion

India's ability to become a global manufacturing powerhouse depends on moving beyond assembly-led growth towards indigenous technology, domestic value addition and resilient supply chains. Indigenisation of critical capital goods is therefore vital to achieving Atmanirbhar Bharat and long-term strategic autonomy.

Drishti Mains Question:

Indigenisation of capital goods is essential for transitioning India from an assembly-based manufacturing economy to an innovation-driven industrial powerhouse. Discuss.

Frequently Asked Questions (FAQs)

1. What is the significance of DPIIT's 100-item indigenisation initiative?
It aims to promote domestic manufacturing of 100 critical intermediates and capital goods, reduce import dependence and strengthen India's industrial supply-chain resilience.

2. What is KABIL and what is its role in India's mineral security?
Khanij Bidesh India Ltd. (KABIL) is a joint venture of three public-sector companies established to identify, acquire and develop overseas critical mineral assets to secure India's mineral supply chains.

3. What is the DESH Bill, 2022?
The proposed Development of Enterprise and Service Hubs (DESH) Bill, 2022 seeks to replace the SEZ Act, 2005, with a broader framework for developing manufacturing and service hubs.

4. What is the significance of the Minerals Security Partnership (MSP) for India?
The MSP is an international partnership aimed at diversifying and securing critical mineral supply chains through cooperation in exploration, extraction, processing and recycling.

5. How does an inverted duty structure affect domestic manufacturing?
It occurs when import duties on raw materials and intermediate inputs exceed those on finished products, raising domestic production costs and discouraging local manufacturing.

UPSC Civil Services Examination, Previous Year Question (PYQ)

Q. With the waning of globalization, the post-Cold War world is becoming a site of sovereign nationalism." Elucidate. **(2025)**

Q. 'What introduces friction into the ties between India and the United States is that Washington is still unable to find for India a position in its global strategy, which would satisfy India's National self-esteem and ambitions'. Explain with suitable examples. **(2019)**

Q. The China-Pakistan Economic Corridor (CPEC) is viewed as a cardinal subset of China's larger 'One Belt One Road' initiative. Give a brief description of CPEC and enumerate reasons why India has distanced itself from the same. **(2018)**

India's Fight Against Child Marriage

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Source: PIB

Why in News?

The **Bal Vivah Mukt Bharat (BVMB) campaign** , launched by the **Ministry of Women and Child Development,** is advancing India's commitment to eliminating child marriage by 2030 through legal enforcement, community participation and digital monitoring.

What are the Key Facts About Child Marriage?

- **Legal Definition:** Under the **Prohibition of Child Marriage Act (PCMA), 2006 ,** child marriage refers to a marriage in which either party is a child, i.e., a female below 18 years or a male below 21 years of age.
- **Prevalence in India:** According to **NFHS-5 (2019–21)** , approximately 23% of women aged 20–24 years were married before attaining 18 years of age, reflecting the persistence of child marriage despite legal prohibitions.
  - States such as **West Bengal and Bihar** have recorded high prevalence, while Uttar Pradesh accounts for a substantial number of child marriages.
- **Social and Health Consequences:** Child marriage **perpetuates poverty, gender inequality, educational exclusion,** early pregnancies, domestic violence and adverse maternal and child health outcomes, particularly among vulnerable rural and tribal communities.
- **Legal Status of Child Marriage:** Under the PCMA, 2006, child marriages are **generally voidable at the option of the contracting party** who was a child, while marriages involving specified circumstances such as trafficking, coercion or violation of a court injunction are void.

- **Protection under Criminal Law:** Under Section 63 of the **Bharatiya Nyaya Sanhita (BNS), 2023** , sexual intercourse or specified sexual acts by a man with his wife below 18 years of age constitute rape, irrespective of her consent.
  - Such acts may also attract provisions of the **Protection of Children from Sexual Offences (POCSO) Act, 2012,** including aggravated penetrative sexual assault where the statutory aggravating circumstances are satisfied.
- **Rights and Legal Safeguards:** The PCMA, 2006, provides for maintenance, residence and custody arrangements to protect affected women and children, while recognising the legitimacy of children born from child marriages.

How Has the Legal Framework Against Child Marriage Evolved in India?

- **19th-Century Social Reforms:** Social reformers such as **Raja Rammohan Roy , Ishwar Chandra Vidyasagar and Mahatma Jyotirao Phule** advocated reforms concerning women's rights, education and harmful marriage practices, contributing to legislative efforts against child marriage.
- **Age of Consent Act, 1891:** Raised the age of consent for sexual intercourse for girls **from 10 to 12 years,** representing an early legislative intervention to protect minor girls.
- **Child Marriage Restraint Act, 1929 (Sarda Act):** Established the **minimum marriage age at 14 years** for girls and 18 years for boys, marking the first major legislative measure specifically aimed at restraining child marriage.
- **Post-Independence Reforms:**
  - **1948 Amendment:** Raised the minimum marriage age for girls from 14 to 15 years.
  - **1978 Amendment:** Increased the minimum marriage age to 18 years for girls and 21 years for boys.
- **Prohibition of Child Marriage Act, 2006:** Replaced the Child Marriage Restraint Act, 1929 (Sarda Act), shifting the legislative approach from merely restraining child marriages to prohibiting them and strengthening legal protection for victims.
- **Beti Bachao Beti Padhao (2015):** Alongside legislation, initiatives such as **Beti Bachao Beti Padhao** , launched in **2015** , have focused on **girls' education, empowerment, awareness and community participation** to address the social drivers of child marriage.

What Are the Key Provisions of the Prohibition of Child Marriage Act, 2006?

- **Legal Definition:** A male below 21 years or a female below 18 years is considered a child, and a marriage is classified as a child marriage if either party is a child.
- **Legal Status of Child Marriage:** Child marriages are generally voidable at the option of the child party, who can petition the **District Court within two years** of attaining majority.
  - Such marriages are **void ab initio** in cases involving trafficking, force, deceit or immoral purposes.
- **Punishments:** Child marriage-related offences are **cognizable and non-bailable,** attracting up to two years of rigorous imprisonment and/or a **fine of ₹1 lakh for adult males marrying children** and persons involved in solemnising, performing, abetting, promoting or permitting such marriages, including parents and guardians.
  - Women offenders are exempt from imprisonment.

- **Child Marriage Prohibition Officers (CMPOs):** States appoint CMPOs to **prevent child marriages, collect evidence,** raise awareness and report relevant data.
- **Preventive Injunctions:** Magistrates can issue injunctions to prevent impending child marriages, and marriages solemnised in violation of such injunctions are void.

What Are the Key Facts About Bal Vivah Mukt Bharat (BVMB)?

- **Launch and Objective:** Launched on 27th November 2024 by the **Ministry of Women and Child Development (MWCD)** , Bal Vivah Mukt Bharat (Child Marriage Free India) aims to eliminate child marriage across India by 2030, particularly among girls and marginalised communities.
- **Constitutional and Global Alignment:** The initiative is rooted in **Article 21 (Right to Life and Personal Liberty)** and the Prohibition of Child Marriage Act (PCMA), 2006, aligning with **SDG Target 5.3,** which seeks to eliminate child, early and forced marriages by 2030.
- **Supreme Court Framework (2024):** In ***Society for Enlightenment and Voluntary Action & Another v. Union of India & Others ,*** the Supreme Court issued comprehensive directions to States and Union Territories to **strengthen child marriage prevention.**
  - **Child Betrothals:** Emphasised the need for legislative amendments to explicitly prohibit child betrothals, which undermine children's autonomy and may lead to forced marriages.
  - **Dedicated Enforcement:** Directed the appointment of full-time Child Marriage Prohibition Officers (CMPOs) at district/sub-district levels and the establishment of Special Child Marriage Prohibition Units for coordination, monitoring and grievance redressal.
  - **Preventive Measures:** Emphasised multi-sectoral awareness campaigns involving schools, Anganwadis, NGOs and religious leaders, training for police, judiciary, teachers and health workers, technology-enabled reporting, and databases identifying at-risk areas.
- **Integrated Prevention Approach:** BVMB shifts the focus from **punishment alone to prevention, protection and empowerment** , strengthening community participation and technology-enabled interventions.
- **Bal Vivah Mukt Bharat Portal:** A centralised, publicly accessible digital platform developed by the **Ministry of Women and Child Development that provides details of appointed Child Marriage Prohibition Officers (CMPOs**), enables real-time reporting of child marriage cases, and tracks awareness campaigns and preventive actions to strengthen stakeholder participation and monitoring.
- **Priority Districts and Funding:** The Ministry of Women and Child Development (MWCD) fully funds the campaign, prioritising 257 high-burden districts identified through NFHS-5 data, where child marriage prevalence is at or above the national average.
- **Community Mobilisation:** The campaign involves schools, educational institutions and Gram Panchayats, with millions of students and other stakeholders participating in nationwide anti-child-marriage pledge-taking activities.
- **International Cooperation:** **UNICEF** has provided technical support for data-driven interventions and capacity-building workshops for Child Marriage Prohibition Officers (CMPOs) and One Stop Centres (OSCs), complementing India's commitments under the **UN Convention on the Rights of the Child (UNCRC).**
- **Convergence with Existing Initiatives:** The initiative builds on earlier efforts such as **Beti Bachao Beti Padhao (BBBP)** by adopting a more **integrated approach to prevention and response** against child marriage.



What Are the Key Features of the 100-Day Campaign Against Child Marriage?

- **About:** The Ministry of Women and Child Development (MWCD) launched a **100-day special campaign on 4th December 2025** across all States and Union Territories to accelerate efforts against child marriage through focused, month-wise outreach activities.
- **Nationwide Pledge:** The campaign commenced at Vigyan Bhawan, New Delhi, with a synchronised national pledge-taking ceremony reaffirming India's commitment to eliminating child marriage.
- **Child Marriage-Free Village Certificate:** Recognises villages and Gram Panchayats that formally pledge to eliminate child marriage and maintain zero reported cases over a sustained period.
- **Bal Vivah Mukt Bharat Yodha Award:** Aims to recognise the **top 10 performing districts, assessed on reporting efficiency,** prevention success and reduction in child marriage cases, through national recognition, commendation certificates and listing on the BVMB portal.
- **District-Level Monitoring:** State Chief Secretaries were directed to **establish district-level task forces comprising** Child Marriage Prohibition Officers (CMPOs), NGOs and Panchayati Raj Institutions (PRIs), with weekly monitoring and geo-tagged progress reporting through the BVMB portal.
- **Targeted Enforcement During High-Risk Periods:** During Akshaya Tritiya 2025, MWCD directed intensified monitoring to **prevent child marriages during the mass-wedding period** , facilitating interventions through judicial injunctions, community counselling and FIR registration.
- **Multi-Sectoral Convergence:** Integrates efforts across education, health and rural development sectors to strengthen community outreach, prevention and coordinated action against child marriage.

Chhattisgarh: Model for Child Marriage Prevention

- **Balod District:** Became India's **first child marriage-free district** , with **zero reported cases for two consecutive years** across **436 Gram Panchayats and 9 urban local bodies** . Chhattisgarh aims to make the entire state child marriage-free by **2028–29** .
- **Surajpur District:** On **17th September 2025** , **75 village panchayats** were declared **"Child Marriage-Free Panchayats"** after recording zero cases for two consecutive years.

Conclusion

India's fight against child marriage has evolved from 19th-century social reforms to comprehensive legal safeguards and community-driven initiatives. The Bal Vivah Mukt Bharat campaign strengthens prevention through legal enforcement, digital monitoring and grassroots participation, aligning with SDG 5.3 and Viksit Bharat. **Sustained focus on education, empowerment and behavioural change** is essential to eliminate child marriage by 2030 and safeguard children's rights and dignity.

Drishti Mains Practice Question

Child marriage in India reflects the intersection of patriarchal social norms, poverty, educational exclusion and weak enforcement. Discuss.

Frequently Asked Questions (FAQs)

Q1. What is Bal Vivah Mukt Bharat?
It is a nationwide campaign launched by the Ministry of Women and Child Development on 27 November 2024 to prevent and eliminate child marriage through awareness, enforcement and community participation.

Q2. What is the target year for making India child marriage-free?
The campaign aims to make India child marriage-free by 2030, in alignment with SDG Target 5.3.

Q3. What is the legal minimum age of marriage in India?
Under the Prohibition of Child Marriage Act, 2006, the minimum marriage age is 18 years for females and 21 years for males.

Q4. Which was India's first child marriage-free district?
Balod district in Chhattisgarh was declared India's first child marriage-free district in 2025, according to the PIB backgrounder.

Q5. What is the role of Child Marriage Prohibition Officers?
CMPOs are appointed under Section 16 of the PCMA, 2006, to prevent child marriages, collect evidence, create awareness and assist with enforcement.

[Watch Video on YouTube: ▶ https://www.youtube.com/embed/oKA5ytxvrX0]

UPSC Civil Services Examination Previous Year Question (PYQs)

Mains

Examine the main provisions of the National Child Policy and throw light on the status of its implementation. **(2016).**

Customs and traditions suppress reason leading to obscurantism. Do you agree? **(2020)**

Cloud Seeding

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Source: IE

Why in News?

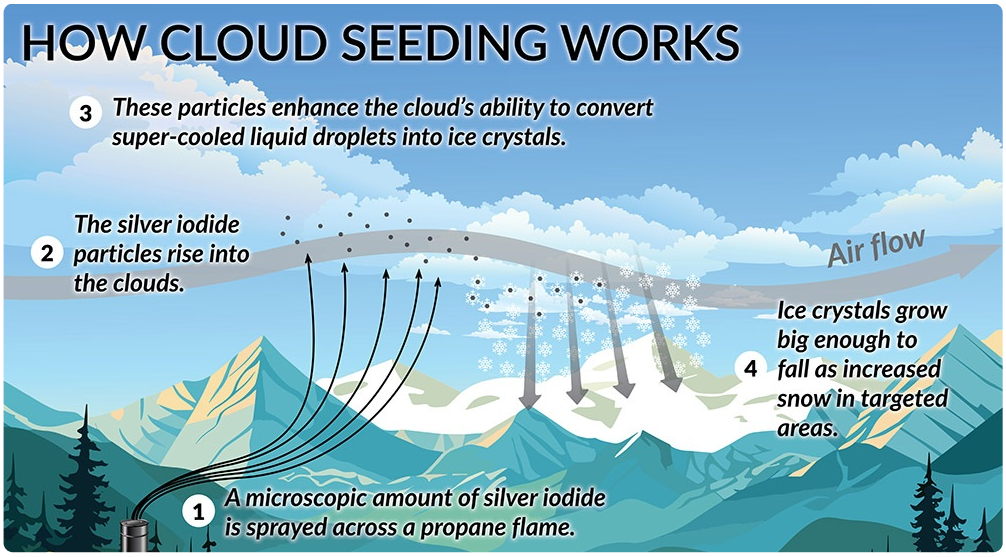
As Delhi and the National Capital Region (NCR) face hazardous **Air Quality Index (AQI)** levels post-Diwali in November 2026, authorities have once again proposed **Cloud Seeding (Artificial Rain) as an emergency mitigation strategy** .

- However, the efficacy of this attempt is being questioned, as the **previous trials conducted in October 2025** yielded negligible rainfall and failed to create any tangible, lasting improvement in the capital's air quality.

What is Cloud Seeding (Artificial Rain)?

- **About:** Cloud seeding is a weather modification technology that **artificially induces or enhances precipitation** by introducing specific chemical substances (Seeding Agents) into existing cloud formations.
  - It aims to accelerate the **natural rain-making process** to manage drought, combat severe air pollution, or suppress hail.
- **Types:**
  - **Glaciogenic Seeding:** Uses ice-nucleating agents such as silver iodide (AgI) to promote ice-crystal formation in suitable cold or supercooled clouds, enhancing snowfall or rainfall.
  - **Hygroscopic Seeding:** Introduces moisture-attracting particles such as **sodium chloride (NaCl)** into suitable warm clouds to encourage droplet growth through collision and coalescence, potentially increasing rainfall.

- **Mechanism of Action:**
  - **Naturally:** Water vapour condenses onto **cloud condensation nuclei (CCN),** such as dust, sea salt, and smoke particles, forming tiny cloud droplets.
    - These tiny droplets grow through the condensation of additional water vapour and collision and coalescence with other droplets.
    - When droplets become sufficiently large and heavy, they can no **longer remain suspended in the atmosphere and fall as precipitation (rain).**
    - In colder clouds, water droplets may freeze or water vapour may **deposit directly onto ice nuclei, forming ice crystals** . These crystals grow and **eventually fall as snow or melt into rain.**
  - **In Cloud Seeding:** Dispersed chemical agents (delivered via aircraft, drones, or ground generators) act as **artificial Cloud Condensation Nuclei (CCN) or ice nuclei.**
    - They provide a **dense, highly effective base** for water vapor to condense and freeze around, **accelerating the formation of water droplets** or ice crystals. Once these droplets become heavy enough, they fall to the ground as artificial rain or snow.



- **Impact on Air Pollution:**
  - **Washout Effect (Coagulation):** As artificial rain falls, the droplets attract and intercept suspended particulate matter (**PM2.5** and **PM10**), soot, sulfates, and toxic **aerosols.**
  - **Settling:** The precipitation physically washes these pollutants out of the atmosphere and deposits them on the ground, leading to a temporary but rapid improvement in the **AQI and surface visibility.**
- **Meteorological Conditions:**
  - **Pre-existing Clouds:** Cloud seeding cannot generate rain from clear skies; it strictly requires existing cloud cover.
  - **Cloud Depth and Moisture:** The target clouds must have sufficient vertical depth and adequate **liquid water content (typically 15-20% humidity in the targeted cloud layer).**
  - **Synoptic Systems:** In regions like North India during winter, cloud seeding relies on the presence of **Western Disturbances** —extratropical storms originating in the **Mediterranean** that bring necessary moisture and cloud formations.
- **Limitations and Challenges of Cloud Seeding:**
  - **Short-Term Relief:** It only addresses the symptoms of air pollution, not the source. An IIT Delhi study found that even after substantial natural rainfall, **pollutant concentrations generally return to pre-rainfall levels within 1–5 days due to continued emissions** .

        - It also highlighted a possible increase in **ground-level ozone concentrations following rainfall.**
    - **Meteorological Dependency:** Success is entirely contingent on unpredictable weather variables.
        - Over Delhi, **winter clouds are often too dry** or shallow to yield significant rainfall even when seeded.
    - **High Capital Cost:** Deploying specialized aircraft equipped with **pyrotechnic flares** and aviation fuel is highly resource-intensive (costing crores for a limited coverage area)
- **Ecological and Ethical Concerns:**
    - **Toxicity and Bioaccumulation:** Prolonged use of silver iodide can lead to the **accumulation of heavy metals in local soils and water bodies,** potentially harming aquatic life and agricultural ecosystems.
    - **Rain Theft:** Altering weather patterns in one micro-region might artificially **deprive neighboring, naturally drought-prone areas of their expected precipitation.**
    - **Carbon Footprint:** Using **dry ice (carbon dioxide)** and flying aviation-fuel-heavy aircraft directly contribute to **greenhouse gas emissions.**

Global Examples of Cloud Seeding Programmes

- **United Arab Emirates (UAE):** Since the 1990s, the UAE has conducted around 1,000 hours of cloud-seeding flights annually to enhance desert rainfall and replenish groundwater.
- **Australia:** Cloud seeding is used to augment water supplies and hydropower generation, particularly in Tasmania and the Snowy Mountains.
- **United States:** Western states such as California, Colorado and Texas use cloud seeding to increase mountain snowpack, supporting reservoirs, irrigation and drought management.
- **China:** Operates one of the world's largest weather-modification programmes using aircraft, drones and artillery for artificial rainfall, drought relief and hail suppression

Alternative Solutions (Beyond Cloud Seeding):

- **Wind Augmentation Purifying Unit (WAYU):** Developed by the Council of Scientific and Industrial Research - National Environmental Engineering Research Institute (**CSIR-NEERI**), these devices are deployed at high-traffic intersections.
    - They work on the principles of **wind generation** and active pollutant removal, utilizing particulate filters and activated carbon/UV systems to neutralize **Carbon Monoxide (CO) and Volatile Organic Compounds (VOCs).**
- **Smog Towers and Downdraft Filtration:** Large-scale outdoor air purification structures (like those tested in Connaught Place, Delhi) that employ downdraft or updraft airflow technology.
    - They pull in polluted air from the top, pass it through thousands of macro and **micro- HEPA (High-Efficiency Particulate Air) or electrostatic filters,** and release clean air at the bottom to create localized clean air zones.
- **Photocatalytic Infrastructure (Smog-Eating Paints):** The application of paints or building coatings infused with **Titanium Dioxide (TiO2)** on urban infrastructure.
    - When exposed to sunlight (UV radiation), these photocatalytic materials trigger a chemical reaction

that breaks down **toxic Nitrogen Oxides (NOx) from vehicle exhaust into harmless nitrate salts,** which are then washed away by natural rain.

- **Anti-Smog Guns and Mist Cannons:** A targeted, ground-level technological intervention that **mimics the washout effect of artificial rain.**
  - These machines spray atomized water droplets **(typically 50-100 microns in size) into the air at high velocities to coagulate** and settle suspended particulate matter (PM2.5 and PM10) around construction sites and mining areas.
- **Pusa Decomposer:** A highly effective biotech alternative to agricultural fires, developed by the **ICAR** .
  - It is a **microbial consortium** (fungi-based bio-enzyme liquid) sprayed onto harvested fields that rapidly accelerates the decomposition of paddy stubble, turning it into organic manure within 15-20 days without the need for burning.
- **AI-Driven Smart Traffic Management:** Implementing **Artificial Intelligence (AI) and Internet of Things (IoT)** synchronized traffic grids to eliminate vehicle idling at red lights.
  - By optimizing traffic flow through real-time data, urban centers can drastically cut down localised concentrations of **tailpipe emissions (NOx and Carbon Monoxide)** at major junctions.
- **Urban Greening:** Developing dense urban forests (**Miyawaki method**) and green corridors to act as natural dust sinks.
- **In-Situ Stubble Management:** Subsidizing and deploying machinery like **Happy Seeders** and Super SMS to prevent agricultural biomass burning in Punjab, Haryana, and UP.
- **Clean Mobility Transition:** Accelerating the shift to **Electric Vehicles (EVs)** through government subsidies and expanding public transit networks to reduce tailpipe emissions.

Frequently Asked Questions (FAQs)

1. What is cloud seeding and which substances are commonly used?
Cloud seeding is a weather modification technique that enhances precipitation by introducing seeding agents such as silver iodide, potassium iodide, sodium chloride and dry ice into existing clouds.

2. What are Cloud Condensation Nuclei (CCN)?
CCNs are tiny atmospheric particles, such as dust, sea salt and smoke, that provide surfaces for water vapour to condense and form cloud droplets.

3. Why is cloud seeding dependent on Western Disturbances in Delhi-NCR?
Western Disturbances bring moisture-bearing clouds and favourable atmospheric conditions during winter, potentially supporting cloud-seeding operations. However, cloud seeding cannot produce rainfall in clear skies.

4. What is the role of CSIR-NEERI's WAYU in controlling air pollution?
The Wind Augmentation Purifying Unit (WAYU) uses wind generation and pollutant-removal mechanisms to reduce local concentrations of particulate matter and gaseous pollutants, particularly at traffic intersections.

5. How does the Pusa Decomposer help control air pollution?
Developed by ICAR, Pusa Decomposer is a microbial consortium that accelerates the decomposition of paddy stubble into organic manure, providing an alternative to crop residue burning.

[Watch Video on YouTube: ▶ **https://www.youtube.com/embed/86MozHiK7zE**]

UPSC Civil Services Examination, Previous Year Questions (PYQs)

Prelims

Q. Artificial way of causing rainfall to reduce air pollution makes use of (2025)

(A) silver iodide and potassium iodide

(B) silver nitrate and potassium iodide

(C) silver iodide and potassium nitrate

(D) silver nitrate and potassium chloride

Ans: A

Q. In the context of which of the following do some scientists suggest the use of cirrus cloud thinning technique and the injection of sulphate aerosol into the stratosphere? (2019)

(a) Creating the artificial rains in some regions

(b) Reducing the frequency and intensity of tropical cyclones

(c) Reducing the adverse effects of solar wind on the Earth

(d) Reducing the global warming

Ans: (d)

Siddis: Gujarat's African-Origin Tribal Community

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Source: IE

The historic achievement of **Khushbu Rajakbhai Darjada,** a 15-year-old judoka who became the **first Siddi athlete to win a Youth Olympics medal** and only the **second Indian judoka** to achieve this feat, has brought renewed attention to the Siddi community, a tribal community of African descent primarily residing in Gujarat.

- **Origins and Arrival:** The Siddis trace their genetic ancestry to **East Africa** .
  - There is no single definitive record of their arrival in India. Some historical accounts (like the **1931 Census**) suggest they were brought as slaves by **Portuguese colonists** in the 17th century, or by Central/West Asian invaders in the 15th century.
  - Community folklore suggests they were brought over 700 years ago by the **Nawabs of Junagadh** to serve as bodyguards (*angrakshaks*), strongmen, and caretakers of captive lions.
- **Demographics and Geography:**
  - **Population:** They are a small community, as per a 2025 government survey, the population of **Siddis in Gujarat classified as a Particularly Vulnerable Tribal Group (PVTG) is 12,959** .

- **Location:** Their population in Gujarat is heavily concentrated in the Saurashtra region, specifically in 18 villages of **Talala taluka in Gir Somnath district** . Jambur village is often referred to as **"Mini Africa."**
    - Members also reside in Maharashtra, Goa, Diu, and Daman.
- **Cultural Assimilation:** Despite their African physical features, Siddis are entirely assimilated into Gujarati culture. They speak Gujarati with a Kathiawadi accent, eat local food, and dress like native Gujaratis.
    - They are predominantly **Muslim** , with their religious beliefs heavily influenced by Sufism, while also incorporating some regional Hindu elements.
    - They revere **Baba Gor** , a 15th- or 16th-century African Muslim saint.
- **Constitutional Status:** Siddis residing in the **Saurashtra region** were declared a **Scheduled Tribe (ST)** in 1956.
    - In 1982, the Government of India classified them as a **PVTG** (formerly known as a "primitive tribal group").
    - However, Siddis living outside the **Saurashtra region** (e.g., in Bharuch) are not classified as STs; they fall under the **Socially and Economically Backward Community (SEBC) category.**
- **Livelihood and Lifestyle:** Due to their proximity to the **Gir National Park** , many Siddis work as forest guides, drivers, and laborers.
    - Often called *Siddi Badshah* (carefree kings), they are known for their simple, free-spirited nature.
    - **Dance:** The Siddis are known for Dhamal, a traditional, energetic folk dance of African origin, performed by men and featuring animal-like movements and calls. Women participate in other traditional Siddi dances.
- **Sporting Prowess:** Considered natural athletes, many Siddi youths excel in sports, particularly martial arts and athletics, leading to their selection in national sports camps, though adapting to strict discipline remains a cultural challenge for some.



Read more: Siddi Community

SC on States' Power to Levy Surcharge on Sales Tax

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Source: IE

A seven-judge Constitution Bench of the Supreme Court in the case ***Arjun Flour Mills v. State of Orissa (2026)*** has resolved a 30-year-old legal dispute by ruling that **states have the constitutional authority to levy a surcharge calculated on sales tax** , declaring it a valid "additional sales tax" rather than an illegal "tax on tax."

- **The 'Tax on Tax' Argument:** The dispute originated from the ***Arjun Flour Mills v State of Orissa* case (1994)** , which challenged an amendment to the **Orissa Sales Tax Act, 1947** .
  - The law imposed an **additional tax (surcharge)** on dealers whose annual turnover crossed specific thresholds (Rs 10 lakh and Rs 1 crore).
  - Dealers argued that since the surcharge was calculated based on the tax payable (and not directly on the turnover), it was an impermissible **"tax on tax"** .
  - They contended it acted like a tax on income, pushing it outside the state's legislative competence under **Entry 54 of List II (State List of the Seventh Schedule)** , which only allows states to tax the "sale or purchase of goods."
- **Supreme Court's Verdict** : The Court rejected the dealers' argument. It held that the **subject of the levy remains the sale or purchase of goods** . Calculating the surcharge as a percentage of the tax payable is merely an "arithmetical route" to determine the final amount, making it a perfectly valid **additional sales tax** .
  - The Bench clarified that the surcharge is tied to a **dealer's turnover** , not their profit. Because it must be **paid even if a dealer suffers a financial loss, it cannot be legally categorized as an income tax.**
- **Resolving Past Legal Conflicts:** For years, dealers relied on the ***India Cement Ltd. v. State of Tamil Nadu (1990)*** ruling, which had classified mining royalty as a tax to argue against taxes calculated on other taxes.
  - This conflict was recently eliminated by a **nine-judge bench in the *Mineral Area Development Authority v. SAIL (2024)*** case, which definitively ruled that **"royalty is not a tax"** but a contractual payment.
  - The Supreme Court upheld the ***1974 judgment in S. Kodar v. State of Kerala,*** affirming that states possess the constitutional authority to impose an additional sales tax on dealers exceeding a specific annual turnover threshold.
- **Surcharge vs. Cess** : A **surcharge is an additional tax** on an existing tax, and its revenue goes directly into the **Consolidated Fund** for general government use.
  - A **cess** , however, is collected for a **specific purpose** (e.g., Education Cess) and must be transferred to a dedicated fund.
- **Post-GST Taxation Powers:** The **101 st Constitutional Amendment Act, 2016, (Goods and Services Tax)** narrowed Entry 54 of the State List to sales taxes on petroleum crude, high-speed diesel, petrol, natural gas, aviation turbine fuel and alcoholic liquor for human consumption.
  - **Article 246A** separately empowers Parliament and State Legislatures to make laws relating to GST, subject to Parliament's exclusive power over inter-state supplies under **Article 269A** .

Read more: Concerns Over Cess and Surcharges in India

Green Firecrackers

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Source: TH

The **Supreme Court of India** recently permitted the limited use of **green firecrackers in Delhi-NCR during Deepavali** , highlighting their emission-reduction technologies and environmental implications.

- **About:** Green firecrackers are specially designed low-emission fireworks developed by the **CSIR-National Environmental Engineering Research Institute (CSIR-NEERI), Nagpur** , to produce fewer pollutants than conventional firecrackers.
- **Emission Reduction:** Green firecrackers must achieve either:
  - **At least 30% lower particulate matter (PM) emissions** , or
  - **At least 20% lower PM emissions and 10% lower gaseous emissions** , including sulphur dioxide (SO_2) and nitrogen dioxide (NO_2), compared with conventional firecrackers of the same category.
- **Emission-Control Mechanism:** They use modified **fuel-oxidiser compositions,** dust-suppressing additives and reduced quantities of certain metals to improve combustion efficiency and limit pollutant formation.
  - Their formulations **exclude banned barium salts** and may use reduced or substituted raw materials.
- **Types of Green Firecrackers:**
  - **SWAS (Safe Water Releaser):** Releases water during combustion, helping suppress particulate matter emissions.
  - **STAR (Safe Thermite Cracker):** Uses a thermite-based composition to improve combustion efficiency and reduce emissions.
  - **SAFAL (Safe Minimal Aluminium):** Contains reduced aluminium content to lower emissions.
- **Certification:** CSIR-NEERI conducts emission testing, while the **Petroleum and Explosives Safety Organisation (PESO)** provides regulatory approval.
- **Residual Pollution:** Despite reduced emissions, green firecrackers still release **fine and ultrafine particulate matter, harmful gases and other combustion by-products** , posing respiratory and cardiovascular health risks.
- **Noise and Air Quality Impact:** Green firecrackers also generate noise, while their overall impact on air quality depends on the **number of firecrackers used, meteorological conditions and existing pollution levels** .

Read More: Green Crackers

REACH Becomes 1st Indian Civil Society NGO to Win Kochon Prize 2026

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Source: TH

The **Resource Group for Education and Advocacy for Community Health (REACH)** , a Chennai-based Indian non-profit organisation working to strengthen **tuberculosis (TB)** control, received the **Kochon Prize 2026** , becoming the **first Indian civil society non-profit organisation** to receive the award.

- **About Kochon Prize:** An international award instituted in **2006** , presented by the **Kochon Foundation, an independent South Korean non-profit foundation, in partnership with the Stop TB Partnership** , recognising outstanding contributions to TB prevention, care and control.
- **2026 Theme:** Integration of TB services with other healthcare services to improve early diagnosis and treatment.
- **Awardees:** The prize was jointly awarded to **REACH (India), Centre for Health Solutions (Kenya), and Marshall Islands TB Control Program** .
- **Previous Indian Recipients:** L.S. Chauhan (2006), head of India's Revised National Tuberculosis Control Programme; Indian Council of Medical Research (ICMR, 2017); and Molbio Diagnostics (2024).
- **About REACH:** A **women-led Indian non-profit organisation** promoting person-centred, integrated TB care through community participation, patient advocacy and public-private partnerships.
- Addresses social, psychological, family-related, mental health and economic barriers to TB treatment alongside clinical needs.
- Works with **community health workers, TB Nanbans and TB Champions** to strengthen patient support and equitable healthcare access.

Read More: World Tuberculosis Day 2026